Ex: 14

Code using RAW Sockets to implement packet sniffing

Aim:

To write a code using RAW Sockets to implement packet sniffing.

Code:

```
from scapy.all import sniff
from scapy.layers.inet import TCP, IP, UDP, ICMP

def packet_callback(packet):
    if IP in Packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        dst_ip = ip_layer.dst
    protocol_name = ""
    if protocol == 1:
        protocol_name = "ICMP"
    elif Protocol == 6:
        protocol_name = "TCP"
    elif Protocol == 7:
        protocol_name = "UDP"
    else
        protocol_name = "unknown Protocol"
    print(f"Protocol: {Protocol_name}")
    print(f"Source IP: {src_ip}")
    print(f"Destination IP: {dst_ip}")
    print("_" * 50)
sniff(iface = "Wi-fi", prn = packet_callback,
      filter = "ip", store = 0)
```

Output :

Protocol : TCP
Source IP : 57.155.141.117
Destination IP : 192.168.1.6

Protocol : TCP
Source IP : 192.168.1.6
Destination IP : 57.155.141.117

Protocol : TCP
Source IP : 192.168.1.6
Destination IP : 57.155.141.117

Protocol : TCP
Source IP : 192.168.1.6
Destination IP : 57.155.141.117